

PDF Generation Automation

Description

This project retrieves data Google Sheets and creates a professional PDF through Gotenberg. Additionally, it automates sending Daily, Weekly, or Monthly, or combination of Sales Report email through Gmail.

Setup

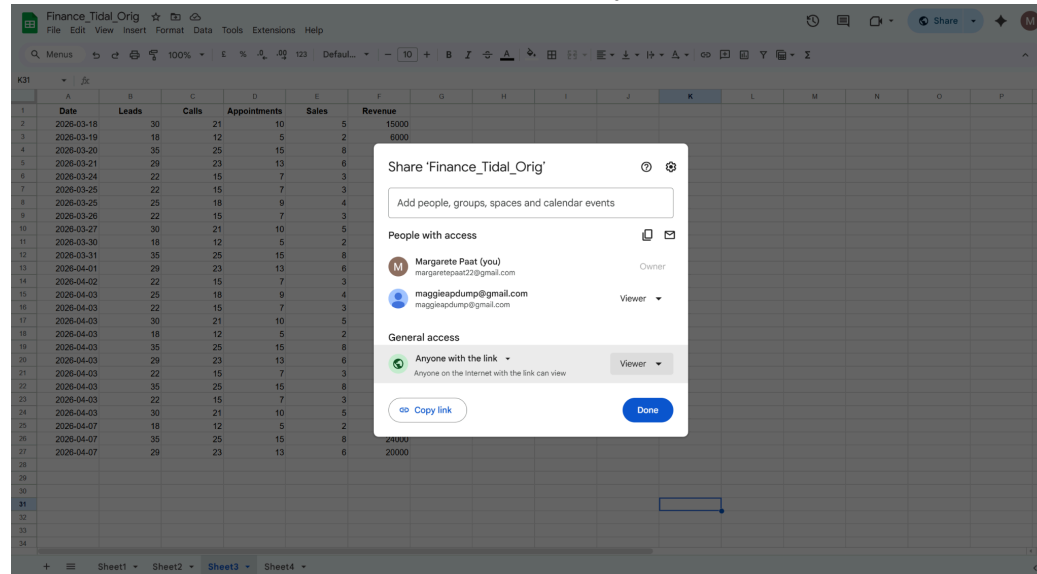
Prerequisites

n8n	Self-hosted or Sign-up to https://n8n.partnerlinks.io/q3whjtr3e9f9
Docker Desktop	1. Create folder where you're gonna store the data and workflows under n8n. Open terminal (or PowerShell/Command Prompt) and run: • <code>mkdir n8n-docker-setup && cd n8n-docker-setup</code> 2. Install Docker • <code>sudo apt update && sudo apt install docker.io docker-compose-v2 -y</code> 3. Create the data directories and configuration files. Run these commands inside project folder: • <code>mkdir n8n_data postgres_data</code> 4. Automatically create the .env and yaml. Note that you will need to configure the necessary details here including APIs and credentials. • <code>touch .env docker-compose.yaml</code> 5. Open the main folder and inside it is the <code>docker-compose.yaml</code>. Add the code below on that file. Configure the <code>.env</code> as well.

	 <pre>Run Service gotenberg: image: gotenberg/gotenberg:8 restart: always ports: - "3010:3000" networks: - n8n_net</pre>
Gotenberg	1. For self-hosted, use Gotenberg as a free and unlimited PDF generator. a. Open Docker Desktop app b. Open command prompt. Set up Gotenberg by running it on command prompt and check on Docker Desktop for confirmation. • <code>docker run --rm -p 3010:3000 gotenberg/gotenberg:8</code> c. When everything is good, start by going to terminal and run • <code>docker start gotenberg</code> • <code>docker ps</code> d. When the port for Gotenberg is already showing, it's working. e. Paste the ngrok link on browser to start creating workflows.

Google Sheets Document

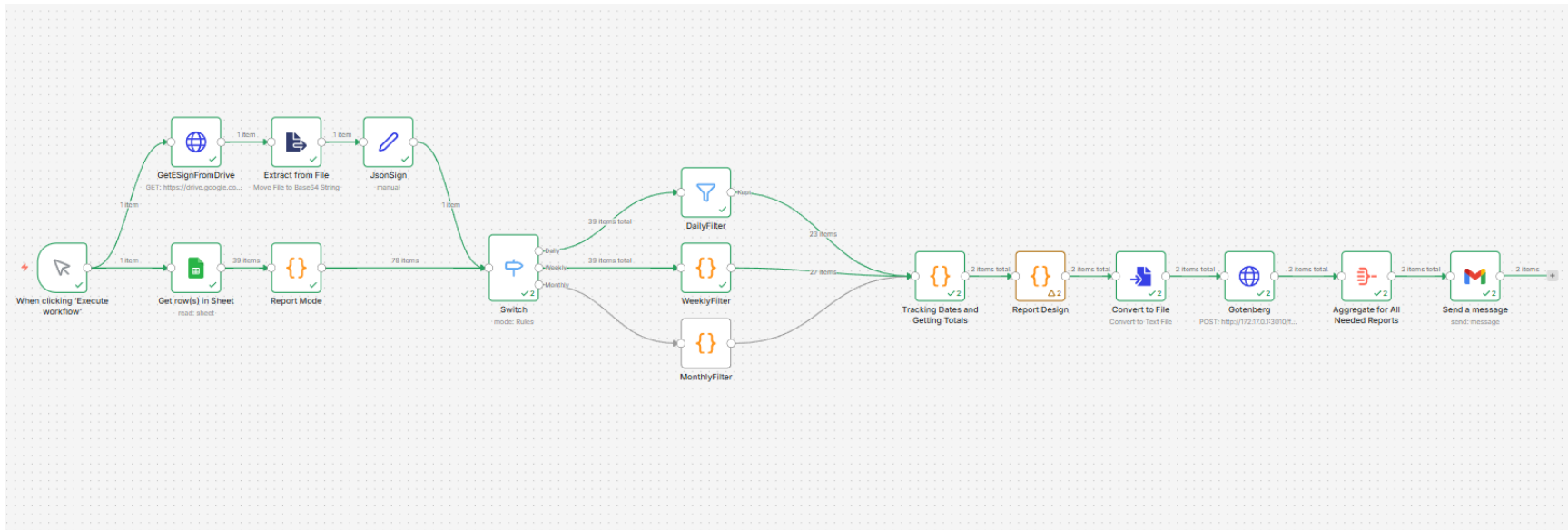
1. On your Sheets, this is where the data is being retrieved.
2. Make sure it is not restricted, and shared with anyone with the link to access.



Gmail

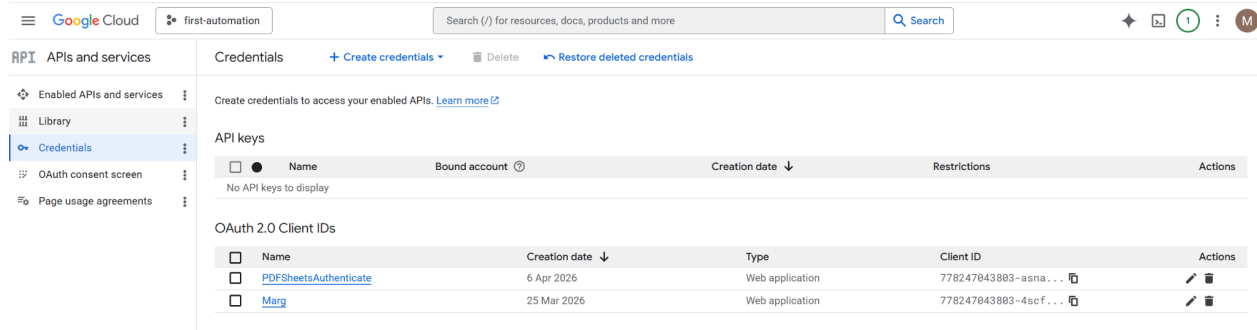
1. Make Gmail address ready because this is where the PDF will be sent for testing
 - a. Save email address and password to be used as credentials later.

n8n Workflow



n8n Credentials Setup

Gmail OAuth2 API	1. Open the Google Cloud Console through https://console.cloud.google.com/ 2. Set up Credentials on the Menu a. Go to APIs and Services > Credentials > OAuth client ID, and get both the Client ID and Client secret to be pasted in Sheets node. b. Add the OAuth Redirect URL in Cloud Console's Authorised redirect URLs indicated in the n8n Sheets node.
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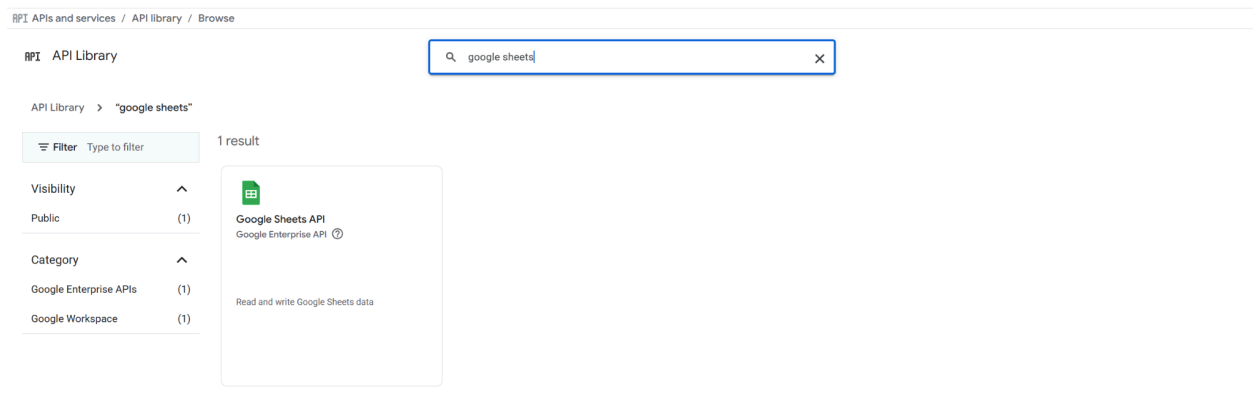


The screenshot shows the Google Cloud console interface. The left sidebar has a menu with 'APIs and services' selected. The main content area is titled 'Credentials' and includes a '+ Create credentials' button. Below this, there are sections for 'API keys' (showing 'No API keys to display') and 'OAuth 2.0 Client IDs'. The 'OAuth 2.0 Client IDs' section contains a table with the following data:

☐	Name	Creation date ↓	Type	Client ID	Actions
☐	PDFSheetsAuthenticate	6 Apr 2026	Web application	778247043803-asna...	
☐	Marg	25 Mar 2026	Web application	778247043803-4scf...	

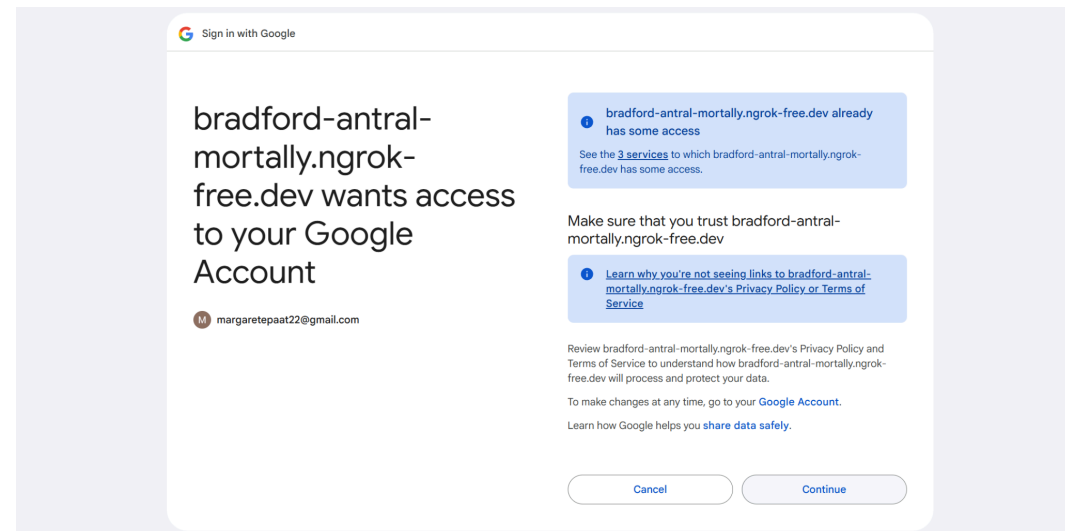
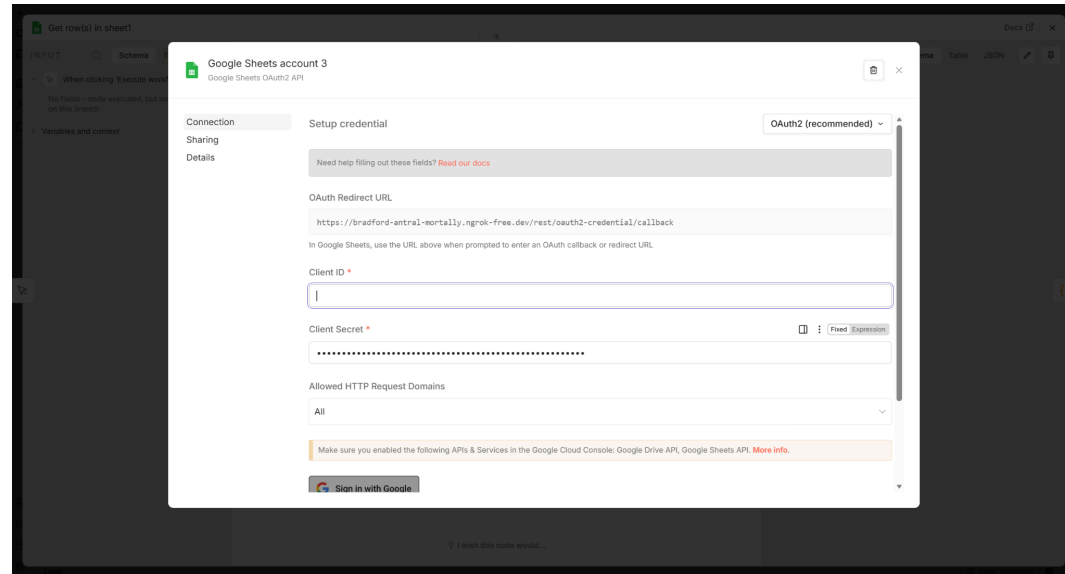
3. Set up API

- Go to APIs and Services > Enabled API and Services.
- Make sure **Google Sheets API** and **Google Cloud APIs** are enabled



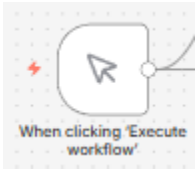
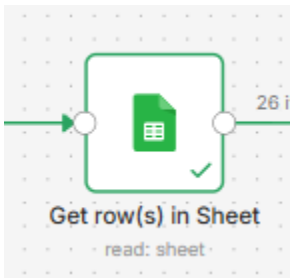
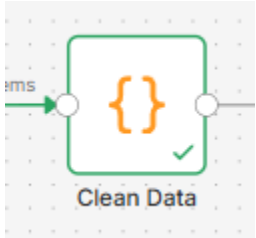
The screenshot shows the Google Cloud API Library search results for 'google sheets'. The search bar at the top contains 'google sheets'. Below the search bar, the results are filtered to show 1 result. The result is a card for the 'Google Sheets API' (Google Enterprise API) with a description 'Read and write Google Sheets data'. On the left side of the results, there are filter options: 'Visibility' (Public, 1), 'Category' (Google Enterprise APIs, 1; Google Workspace, 1).

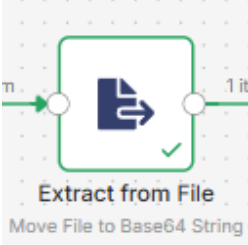
4. Save the details and sign in with a Gmail account. Tick the checkboxes based on needed information.

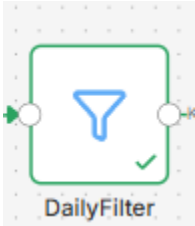
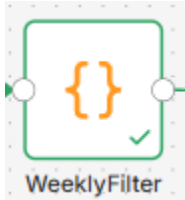
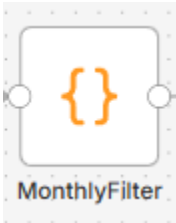


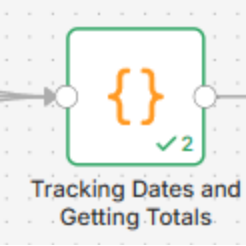
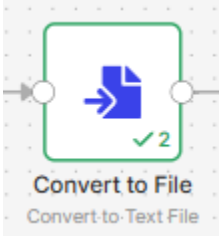
5. Do the same with the Gmail node.

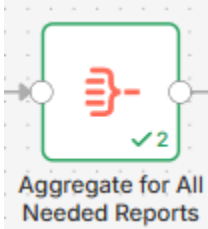
n8n Workflow Setup

Create a New Workflow	
Node	Configuration
1. Manual Trigger 	→ On the empty canvas, click the + button and Search for Manual Trigger → This starts the workflow, so add additional nodes through clicking + button
2.1 Get rows in Sheet 	→ Click the Google Sheets node, click Get row(s) in sheet → Set up credentials, then sign in with a Gmail account → Under Resource, choose Sheet Within Document → Choose Get Row(s) as the Operation → The Sheet name will appear when From list is chosen → Choose the specific sheet name and choose AND under Combine Filters
2.2 Clean Data 	→ Get current UTC time → Identifies the type of report (Daily, Weekly, Monthly, or combination)
2.3 Get e-signature	→ Use the Method GET → Paste in the URL the photo of the e-signature found in Drive → Response Format: File → Put Output in Field: data

 GetESignFromDrive ET: https://drive.google.co...	
2.4 Extract from File  Extract from File Move File to Base64 String	→ Operation: <i>Move File to Base64 String</i> → Input Binary Field: <i>data</i> → Destination Output Field: <i>signatureBase64</i>
2.5 Edit Fields1  Edit Fields1 manual	→ Mode: Manual Mapping → Fields to Set (String) ◆ Name: signatureBase64 ◆ Value (expression): {{ \$json.signatureBase64 }}
3. Switch  Switch	→ Results go to three output outlets → Mode: Rules → Routing Rules ◆ Value1 (expression): {{ \$json.reportMode }} ◆ Value2 (expression): Daily ◆ Rename Output Toggle On: Daily ◆ Value1 (expression): {{ \$json.reportMode }} ◆ Value2 (expression): Weekly ◆ Rename Output Toggle On: Weekly

	◆ Value1 (expression): {{ \$json.reportMode }} ◆ Value2 (expression): Monthly ◆ Rename Output Toggle On: Monthly
4. Daily Filtering 	Daily report is fetched to be sent. Run every 5PM weekdays.
5. Weekly Filtering 	Weekly report is fetched to be sent. Only run every 5PM Friday.
6. Monthly Filtering 	Monthly report is fetched to be sent. Only run every 5PM end of month (28, 30, 31)
7. Tracking Dates and Sum	Language: JavaScript Mode: Run Once for All Items → Rechecks for empty data

 Tracking Dates and Getting Totals	→ Determine the Report Type (Daily, Weekly, Monthly) → Calculate Totals for every column → Clean up data rows for a clean table dateRange: it identifies range of week or month or date of day Signature: The code looks back to <code>JsonSign</code> to grab the signature image.
8. Design  Report Design	Language: JavaScript Mode: Run Once for All Items → It pulls the totals, rows, and signature from the previous node's output. → It loops through data rows to create table rows and <code>Total</code> row at the bottom. → Has CSS for the logo, colors, and layout to look correct, including a fixed signature placement at the bottom of the page. → There was customization for company name, address, and role of the signor. → It outputs a <code>file.html</code>.
9. Convert to file  Convert to File Convert to Text File	Operation: Convert to File Text Input Field: html Put Output File in Field: data File Name: index.html This keeps the signature inside the file and for technical requirement of Gotenberg
10. Gotenberg  Gotenberg T: http://172.17.0.1:3010	Method: POST URL: <code>http://<ip address>/forms/chromium/convert/html</code> Authentication: None Send Headers: Using Fields Below Name: Gotenberg-Output-Filename Value: <code>{{ \$node["node-that-filters-and-calculates"].json["reportType"] }}_Report</code> Send Body

	Body Content Type: Form-Data Type: n8n Binary File Name: index.html Input Data Field: data Options Include Response Headers and Status: On Response Format: File Put Output in Field: {{ \$node["node-that-filters-and-calculates"].json["reportType"] }}
11. Aggregate  Aggregate for All Needed Reports	Aggregate: All Item Data (Into a Single List) Include: All Fields Options: Include Binaries On This combines all the rows into one single report.
12. Gmail  Send a message	→ Put credential first → Resource: Message → Operation: Send → To: <put gmail address here> → Subject: Sales Report (for example) → Email Type: HTML → Message: put any message here before the pdf → Attachment: {{ \$node["node-that-filters-and-calculates"].json["reportType"] }} This is the final step where the pdf is sent.

Sample PDF Docs

PHILCARE
HEALTHCARE MADE SMARTER

STI Holdings Center
6764 Ayala Avenue,
Makati City

WEEKLY SALES REPORT

March 30 - April 3, 2026

DATE	LEADS	CALLS	APPOINTMENTS	SALES	REVENUE
2026-03-30	18	12	5	2	6,000
2026-03-31	35	25	15	8	24,000
2026-04-01	29	23	13	6	20,000
2026-04-02	22	15	7	3	9,500
2026-04-03	25	18	9	4	12,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	22	15	7	3	9,500
2026-04-03	22	15	7	3	9,500
2026-04-03	25	18	9	4	12,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	29	23	13	6	20,000
2026-04-03	22	15	7	3	9,500

DATE	LEADS	CALLS	APPOINTMENTS	SALES	REVENUE
2026-04-03	35	25	15	8	24,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	35	25	15	8	24,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	29	23	13	6	20,000
TOTAL	715	507	263	126	388,000


JOHN SMITH
SALES OFFICER

PHILCARE
HEALTHCARE MADE SMARTER

STI Holdings Center
6764 Ayala Avenue,
Makati City

DAILY SALES REPORT

April 3, 2026

DATE	LEADS	CALLS	APPOINTMENTS	SALES	REVENUE
2026-04-03	25	18	9	4	12,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	22	15	7	3	9,500
2026-04-03	22	15	7	3	9,500
2026-04-03	25	18	9	4	12,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	29	23	13	6	20,000
2026-04-03	22	15	7	3	9,500
2026-04-03	35	25	15	8	24,000
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	35	25	15	8	24,000

DATE	LEADS	CALLS	APPOINTMENTS	SALES	REVENUE
2026-04-03	22	15	7	3	9,500
2026-04-03	30	21	10	5	15,000
2026-04-03	18	12	5	2	6,000
2026-04-03	35	25	15	8	24,000
2026-04-03	29	23	13	6	20,000
TOTAL	611	432	223	107	328,500


JOHN SMITH
SALES OFFICER

Date: 04/20/2026

Subject: Sales Report

Good day!

The automated sales report has been processed and is attached. Please review the attached file for comprehensive details.

Best regards,

Sales Operations

Thank You!

- Margarete Paat | Tidal Solutions Corp. 2026 -